

MindTrack AI

Mental Health Journal & Emotion Tracker

Project Specification Document

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1 Project Overview

MindTrack AI is an AI-powered mental wellness platform that allows users to maintain a personal journal and track their emotional well-being. Users can record daily thoughts and experiences, while the AI analyzes journal entries to identify emotions, monitor mood patterns, and provide supportive wellness insights.

The project aims to promote emotional awareness and self-reflection among students and young professionals through intelligent journaling and mood tracking.

2 Problem Statement

Mental health challenges such as stress, anxiety, and emotional burnout are increasingly common among students and professionals. Many individuals find it difficult to regularly reflect on their emotional state or identify mood patterns over time.

MindTrack AI provides a safe digital platform where users can document their daily experiences and receive AI-powered emotional insights, helping them better understand their mental well-being.

3 Target Users

- Students
- Working Professionals
- Job Seekers
- Individuals interested in emotional wellness

4 Objectives

- Enable users to maintain daily journal entries.
- Analyze emotions using Artificial Intelligence.
- Track mood trends over time.
- Generate personalized wellness suggestions.
- Promote self-awareness and emotional reflection.

5 MVP Feature List (Phase 1)

The following five features form the Minimum Viable Product for the first release:

1. **User Registration & Login** — Secure authentication via Supabase Auth.
2. **Create / View Journal Entries** — Full CRUD support for personal journal entries.
3. **AI Emotion Detection** — Each entry is analyzed by OpenAI to detect the dominant emotion.

4. **Mood Score Tracking** — A numeric mood score (0–10) is assigned and stored per entry.
5. **Basic Wellness Suggestion** — AI returns a short personalized suggestion based on detected emotion.

6 Core Features ---

6.1 User Management

- User Registration
- User Login
- Secure Authentication using Supabase

6.2 Journal Management

- Create Journal Entry
- View Previous Entries
- Update Entries
- Delete Entries

6.3 AI Features

- Emotion Detection
- Sentiment Analysis (positive / negative / neutral)
- Mood Scoring (scale 0–10)
- Personalized Wellness Suggestions
- Weekly Mood Summary Generation

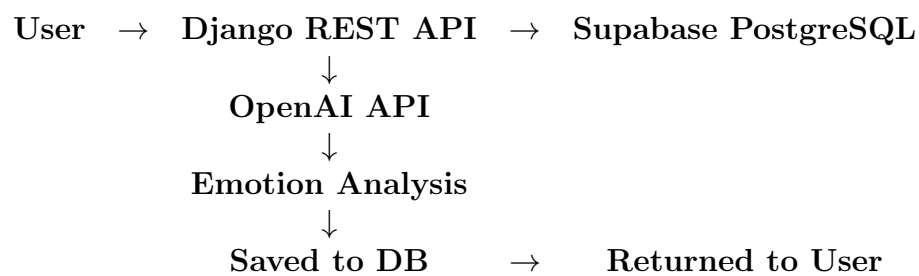
6.4 Analytics

- Mood History Tracking
- Emotion Trends Visualization
- Weekly Wellness Reports

7 Technology Stack

Component	Technology
Backend	Django REST Framework
Database	Supabase PostgreSQL
Authentication	Supabase Auth
AI Integration	OpenAI API (GPT-4)
API Documentation	Swagger / drf-spectacular
Deployment	Railway
Version Control	Git & GitHub

8 System Architecture



9 Data Model

9.1 Users

Field	Type	Notes
id	Integer	Primary Key, Auto-increment
name	String	Full name
email	String	Unique, used for login
password	String	Hashed
created_at	DateTime	Auto set on registration
is_active	Boolean	Default: True

9.2 Journal Entries

Field	Type	Notes
id	Integer	Primary Key
user_id	Integer	Foreign Key → Users
entry_text	Text	Main journal content
mood_score	Integer	0–10 scale
created_at	DateTime	Auto set on creation
updated_at	DateTime	Auto updated on edit

9.3 Emotion Analysis

Field	Type	Notes
id	Integer	Primary Key
entry_id	Integer	Foreign Key → Journal Entries
emotion	String	e.g., happy, sad, anxious
sentiment	String	positive / negative / neutral
mood_score	Integer	0–10 scale
ai_feedback	Text	Wellness suggestion from AI
created_at	DateTime	Auto set on analysis

10 API Endpoints

Method	Endpoint	Description
POST	/api/auth/register/	Register a new user
POST	/api/auth/login/	Login and get token
GET	/api/journals/	List all journal entries
POST	/api/journals/	Create a new entry
GET	/api/journals/{id}/	Retrieve a single entry
PUT	/api/journals/{id}/	Update an entry
DELETE	/api/journals/{id}/	Delete an entry
POST	/api/analyze/{id}/	Trigger AI emotion analysis
GET	/api/mood/history/	Get mood trend data
GET	/api/mood/summary/	Get weekly wellness report

11 Expected Outcomes

- Improved emotional awareness among users.
- AI-assisted journaling experience.
- Personalized wellness insights.
- Easy-to-use mood tracking system.
- Secure and scalable cloud-based platform.

12 Future Enhancements

- Mobile Application (React Native)
- Voice Journal Entries
- Mood Prediction using Machine Learning

- Community Support Features
- Mental Wellness Resources Recommendation

13 Conclusion

MindTrack AI combines journaling, emotional analysis, and artificial intelligence to create a platform that supports mental wellness and self-reflection. The project demonstrates the integration of Django REST Framework, Supabase, OpenAI APIs, and cloud deployment technologies while addressing a socially relevant problem. With a clearly defined MVP, structured data model, and well-documented API endpoints, the project is positioned for successful development and deployment.